C4 System Context (Level 1) — Fire Detection & Autonomous Response

Building operator [Person]
watches viewer and dashboards, triggers test fires and false alarms

Building occupants [Person]
presence drives the occupancy model; evacuation routes are computed for them

watches doors, effects, alert cards [3D viewer]

starts/stops fires, triggers false alarms, clears the viewer [REST control API]

presence sensed [occupancy model]

monitors incidents, latencies, controller health [Grafana]

Other building systems
[Software Systems]
Ventilation, security, energy management (not integrated; share BuildSim)

Fire Detection & Autonomous Response
[Software System]
Tier 1: edge controllers per floor (Raspberry Pi) detect and release doors locally.
Tier 2: server agent confirms with an LLM, coordinates floors, stands the building down.

sensors / actuators [REST]

reads floor plans, door graph, equipment; commands doors; pushes effects and cards [REST]

asks for a verdict on each alert {is_fire, confidence} [HTTP, ~2 s]

readings, decisions, verdicts, heartbeats [MQTT → TimescaleDB]

BuildSim
[Software System]
Building model of LTU A-huset: 950 rooms, 124 fire doors, sprinklers, 3D viewer

LLM inference service
[Software System]
LiteLLM → SGLang, Qwen2.5-14B, self-hosted on the cluster GPU

Observability
[Software Systems]
Grafana dashboards, Phoenix LLM traces, ClearML training runs, TimescaleDB